

ERP Mobile App – Flow Documentation

Flutter-based ERP Application with Strict RBAC

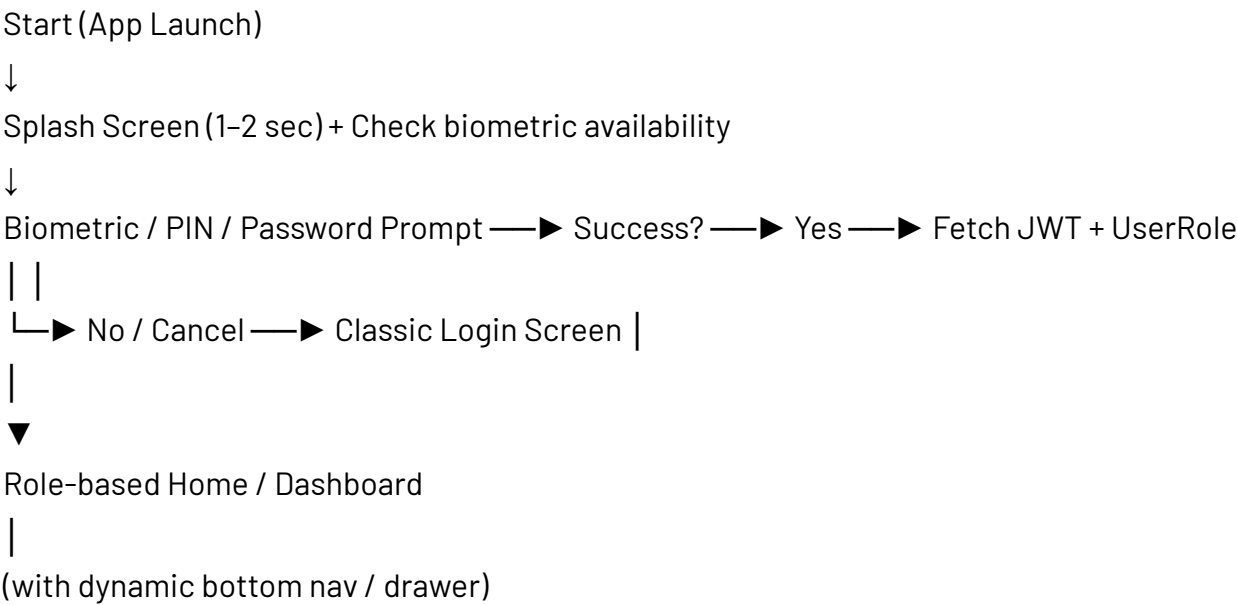
Date: March 03, 2026

1. Document Purpose & Scope

- Map complete user journeys per role
- Show entry points → decisions → screens → outcomes
- Include RBAC gates, success paths, error paths, offline scenarios
- Focus: Finance, HR, Inventory, Sales, Admin + global flows
- Visualization style recommendation: Vertical mobile frames, color-coded paths per role

2. Global Flows (Common to All Users)

2.1 Authentication & Onboarding Flow



Variants:

- First-time → Onboarding carousel (3-5 slides) → Set PIN/biometric
- Biometric fail 3× → Fall back to password
- Token expired → Force re-login
- Offline → Cached last role + warning banner

3. Role-Specific Main Flows

3.1 Executive Role (Operational / Task Executor)

Primary Goal: Perform daily assigned tasks quickly

Main Happy Path – Create Sales Order

Home Dashboard (shows My Tasks widget + Quick Actions)
↓ (tap "New Order")
Sales → Customer Search / Select
↓
Customer Detail / New Customer (mini form)
↓
Add Products (barcode scan or search)
↓
Order Items List + Quantity + Price override (limited)
↓
Review Order → Apply basic discount? (only if allowed)
↓
Submit Order → API call
└─▶ Success ─▶ Order Confirmation Screen + Print/Share button
└─▶ Fail (validation/network) ─▶ Error modal + Save Draft option

Other Key Flows (Executive):

- Inventory → Scan item → Update stock count → Submit count
- HR → My Profile → Submit Leave Request → Form → Submit
- Finance → Submit Expense → Camera upload receipt → Submit

3.2 Manager Role (Approval + Oversight)

Primary Goal: Approve, review team performance, delegate

Main Happy Path – Approve Purchase Order

Home Dashboard (shows Pending Approvals widget + Team KPIs)
↓ (tap Pending PO #12345)
Inventory → Purchase Orders → Pending Tab
↓
PO Detail Screen (read-only + Approve/Reject/Comment buttons)
└─▶ Approve ─▶ Add optional note → Confirm dialog → API approve
| ↓

| Notification sent to requester + procurement team
|
| —► Reject —► Mandatory comment + reason dropdown → Confirm
| ↓
| Status → Rejected + notify requester
|
| —► Comment Only —► Internal note → no status change
text

Other Key Flows (Manager):

- HR → Employee List → Select employee → View Performance → Start Review → Fill form → Submit
- Sales → Sales Analytics → Filter team → Drill-down to orders → Approve special discount request
- Finance → Pending Invoices → Bulk approve / reject

3.3 Admin Role (System Configuration)

Main Happy Path – Create New User

Admin Home (shows System Health + Quick Setup cards)

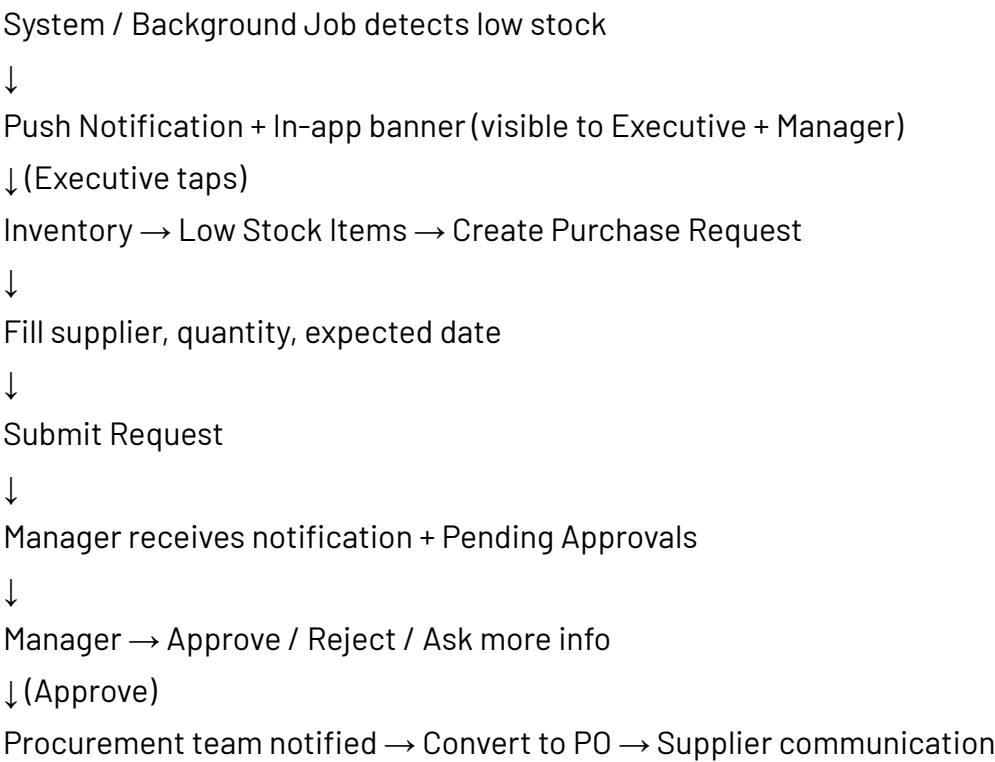
↓
Admin → User Management
↓
User List → + New User button
↓
New User Form
• Full name, email, phone
• Select Role (dropdown: Executive / Manager / Admin / Viewer)
• Assign Modules (multi-select)
• Set temporary password
↓
Save → Validation → API create user
↓
Success → Show generated temp password + Send email toggle

Other Key Flows (Admin only):

- Settings → Theme / Notification global config
- Audit Log → Filter by date / user / module → Export CSV

4. Critical Cross-Module Flows with RBAC Gates

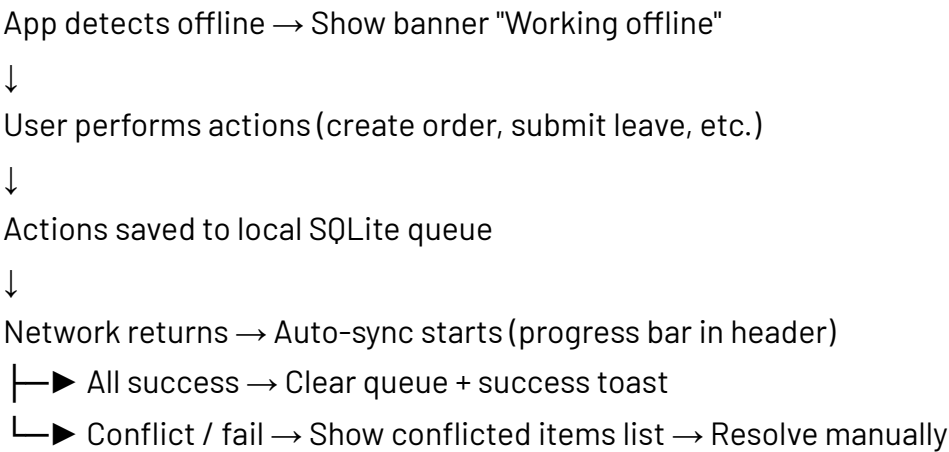
4.1 Low Stock Alert → Purchase Request → Approval Chain



RBAC Summary Table

Action	Executive	Manager	Admin
View low stock alert	Yes	Yes	Yes
Create purchase request	Yes	Yes	No
Approve purchase request	No	Yes	No
View full approval history	No	Yes	Yes

4.2 Offline → Online Sync Flow (All Roles)



5. Error & Edge Case Flows (Global)

- 401 Unauthorized → Redirect to Login + clear cache
- 403 Forbidden (RBAC violation) → Access Denied Screen with "Contact Admin" button
- No Internet – Critical Action → "This action requires internet" modal + Retry button
- Session Timeout → Background → Re-authenticate with biometric if enabled

6. Recommended Visualization Style (2026 Trends)

- Vertical mobile phone frames chained with arrows
- Color-code paths: Executive = blue, Manager = green, Admin = purple
- Decision diamonds for Yes/No branches
- Dashed lines for optional / error paths
- Annotations for: API calls • local storage • notifications
- Include small icons: biometric, dark mode, offline